



ASSESSMENT MATHEMATICS

Psychometric & Archetype Match Formulas

SECURE STEPS ASSESSMENT PLATFORM - TECHNICAL MANUAL

DOCUMENT PURPOSE:

Reference guide for scoring metrics & calibration values.

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May 24, 2026

1. Core Psychometric Dimensions & Normalization

The SecureSteps assessment engine evaluates students across 6 bipolar personality dimensions. Each option selected in the assessment contributes a pre-defined positive or negative weight vector to one or more dimensions. Because the test length is dynamic (incorporating adaptive follow-up slots), raw scores are normalized relative to the maximum possible absolute capacity of the path taken.

decision_style:	Intuitive (negative) vs. Deliberate / Analytical (positive)
energy:	Introverted / Reflective (negative) vs. Extroverted / Expressive (positive)
structure:	Open-ended / Spontaneous (negative) vs. Organized / Structured (positive)
risk:	Cautious / Risk-averse (negative) vs. Bold / Risk-tolerant (positive)
social:	Independent / Solo (negative) vs. Collaborative / Networked (positive)
drive:	Reactive / Standard (negative) vs. Proactive / Build-first (positive)

Normalization Formulas:

Let Q be the set of answered questions. For each question q in Q , the student selects an option o_q .

1. Raw Score calculation: $\text{Raw_dim} = \sum_{q \in Q} (o_q.\text{scores.dim})$
2. Maximum Absolute Capacity: $\text{Max_dim} = \sum_{q \in Q} (\max_{o \in q.\text{options}} |o.\text{scores.dim}|)$
3. Normalized Dimension Score: $\text{Normalized_dim} = \text{round}((\text{Raw_dim} / \text{Max_dim}) * 100)$

This maps every dimension onto a clean $[-100, 100]$ scale, ensuring that unanswered or skipped adaptive questions do not deflate or inflate the final metrics.

2. Archetype Match Formulas & Calibration

The system maps the normalized dimension scores into compatibility match ratings for 6 distinct user archetypes. The archetype with the highest match score is assigned as the student's primary archetype.

The Builder **Match = drive + risk + max(0, -structure)**

Action-oriented; values building fast over rigid rules.

The Strategist **Match = decision_style + structure + max(0, -risk / 2)**

Deliberate and organized; values cautious planning. (Calibrated)

The Connector **Match = social + energy + max(0, -decision_style / 2)**

People-centric; values collaboration and networking.

The Maverick **Match = risk + max(0, -structure) + max(0, drive / 2)**

Risk-tolerant lone wolf; comfortable with high ambiguity.

The Anchor **Match = structure + max(0, -risk) + max(0, -energy / 2)**

Orderly and calm under pressure; resists noisy trends.

The Explorer **Match = max(0, -structure) + drive + max(0, energy / 2)**

Curious polymath; prefers breadth of interest over narrow focus.

Calibration Release Notes (May 24, 2026):

Problem: In testing, a large percentage of students were classified as 'The Strategist'. This occurred because its old formula 'decision_style + structure + max(0, drive / 2)' added a drive boost that was frequently positive (+25 pt average boost on balanced profiles), creating a strong mathematical bias.

Solution: Replaced the drive boost with a cautiousness boost 'max(0, -risk / 2)'. This reflects the psychology of a Strategist (risk-averse/deliberate) and balances the scoring. Balanced positive profiles now tie evenly at 100 points across Builder, Strategist, Connector, and Explorer, eliminating default bias.

3. Match Confidence Score (matchScore)

The report surfaces a final match confidence score (between 70% and 99%) on the cover page. This represents how distinct the user's primary archetype signature is from their overall baseline.

Step-by-Step Match Confidence Algorithm:

1. Absolute Ranking:

Sort the absolute values of the 6 normalized dimension scores in descending order:

$S = \text{sort}([| \text{Normalized_ds} |, | \text{Normalized_en} |, | \text{Normalized_st} |, \dots])$

2. Spread Derivation:

Calculate the spread between the strongest defining indicator and the median indicator:

$\text{Spread} = \max(0, S[0] - S[3])$

3. Base Confidence Score:

$\text{Base} = 78 + \min(18, \text{Spread} * 0.25)$ [Yields a baseline range between 78 and 96]

4. Engagement Adjustment:

Incorporate the user's interaction bandwidth (speed and emoji reactions):

$\text{EngBoost} = (\text{Score_engagement} - 1.5) * 1.5$ [Yields adjustments between -2.25 and +2.25]

5. Clamping:

$\text{Final Match Score} = \text{clamp}(70, 99, \text{round}(\text{Base} + \text{EngBoost}))$

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